



SAFETY DATA SHEETS

Metalaxyl + Propamocarb HCl 25% WP

No.: 112010

Version: 3

Date: March, 2019

SECTION 1: IDENTIFICATION

Product identifier	Metalaxyl 10%+ Propamocarb HCl 15% WP
Other means of identification	N/A
Recommended use	Fungicide
Supplier's details	Zhejiang Heben Pesticide & Chemicals Co., Ltd
Address	Liandun Road, Houjing, Yanjiang Industrial Area, Wenzhou, Zhejiang, China
Telephone No	+86-577-88797730; +86-577-88797721
Fax No.	+86-577-88797739
E-mail	hb-p@hb-p.com
Emergency phone number	+86-532-83889090

SECTION 2: HAZARDS IDENTIFICATION

2.1 GHS classification of the substance or mixture

Physical hazards	None	None
Health hazards	Serious eye damage/eye irritation, Category 2B	H320
Environmental hazards	Acute aquatic toxicity, Category 3	H402
	Chronic aquatic toxicity, Category 3	H412

2.2 GHS label elements, including precautionary statements

Hazard pictograms	No symbol
Signal word	Warning
Hazard statement (s)	
H320	Causes eye irritation.
H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

Prevention statements

P264	Wash hands and face thoroughly after handling.
P273	Avoid release to the environment.



P280

Wear protective gloves/protective clothing.

Response statements

P305 + P 351 + P338

If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P391

Collect spillage.

Storage statements

P403 + P233

Store in the well-ventilated place, keep container tightly closed.

P405

Store locked up.

Disposal statements

P501

Dispose of contents/container in accordance with local regulations.

2.3 Other hazards which do not result in classification

No other hazards known.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Common name, synonyms	Chemical identity	CAS number and other unique identifiers	The concentrations of the ingredients
Propamocarb HCl	propyl [3-(dimethylamino)propyl]carbamate hydrochloride (IUPAC name)	CAS No.: 25606-41-1 EC No.: 247-125-9	15%
Metalaxyl	Methyl N-(2,6-dimethylphenyl)-N-(methoxyacetyl)-DL-alaninate. (IUPAC name)	CAS No.: 57837-19-1; EC No.: 260-979-7	10%
Other non-hazardous ingredients			< 75%

SECTION 4: FIRST-AID MEASURES**4.1 Description of necessary first-aid measures**

Skin: Remove contaminated clothes. Rinse and then wash skin with water and soap. Refer for medical attention.

Eyes: For eye contamination, flush eyes immediately with water. Irrigate each eye continuously with normal saline during transport.

Inhalation: Move affect person to fresh air and keep at rest until recovered. If not breathing, give artificial respiration and get to a doctor.

Ingestion: Do not induce vomiting if the person is conscious. Give glass of water. Get to a doctor.



4.2 Most important symptoms/effects, acute and delayed

No such information is reported.

4.3 Indication of immediate medical attention and special treatment needed, if necessary

No antidote, no special treatment, please treat it symptomatically.

SECTION 5: FIRE-FIGHTING MEASURES

5.1 Suitable extinguishing media

Use dry chemical, carbon dioxide, water spray, and foam.

5.2 Specific hazards arising from the chemical

May produce toxic fumes of nitrogen oxides, chlorides, carbon dioxide and carbon monoxide if burn.

5.3 Special protective equipment for firefighters

Should wear full-protective clothing, and self-contained breathing apparatus. Fight fire from safe distance and protected location. Avoid (reject) fire-fighting water to enter environment.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Wear full protective clothing and self-contained breathing apparatus. Control the spill at its source. Dike area and absorb small spills with materials such as sand, sawdust, Zorb all, or dirt and place in suitable containers for recovery or disposal. Remove all contaminated clothing promptly and wash exposed body areas thoroughly with soap and water immediately after handling. Thoroughly launder clothing before reuse. Do not take clothing home to be laundered.

6.2 Environmental precautions

Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems of any body of water. Keep spills and cleaning run-off out of municipal sewers and open bodies of water.

6.3 Methods and materials for containment and cleaning up

If there is contamination of crops or waterways, advise emergency services or state department of agriculture.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid direct or prolonged contact with skin and eyes. Do not breathe dust. Do not ingest. It is recommended that wear full protective clothing including face mask, face shield and gauntlets, all skin areas should be covered, when handling this product.

Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. And take a bath or wash hands completely with soap after use. Remove contaminated clothing and protective equipment before entering eating areas.

Prevents handling of incompatible substances or mixtures when use this product. Minimize the release of this product to the environment when handling this product.

**7.2 Conditions for safe storage, including any incompatibilities**

Store the material in a well-ventilated, dry, cool, out of light and secure area, out of reach of children and domestic animals, and in sealed original containers. Do not store food, beverages or tobacco products in the storage area. Store this product away from the incompatible materials, explosive atmospheres, corrosive conditions, fire and heat, etc.

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION**8.1 Control parameters**

Contain no substances with occupational exposure limit values.

8.2 Appropriate engineering controls

Use only in an enclosed system. Use local exhaust ventilation. Safety shower. Use explosive dust handling controls.

8.3 Individual protection measures

Industrial hygiene: Remove and wash contaminated clothing promptly. After use and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water.

Personal protective equipment**Respiratory protection:**

Wear respirator with a particle filter mask (protection factor 20) conforming to European Norm EN149FFP3 or EN140P3 or equivalent.

Respiratory protection should only be used to control residual risk of short duration activities, when all reasonably practicable steps have been taken to reduce exposure at source e.g. containment and/or local extract ventilation. Always follow respirator manufacturer's instructions regarding wearing and maintenance.

Protective gloves: rubber gloves;

Eye protection: Wear goggles (conforming to EN166, Field of Use = 5 or equivalent).

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES**9.1 Information on basic physical and chemical properties**

Physical state	Powder
Color	Off-white
Odor	Pungent odor
Melting point / freezing point	Not determined for mixture
Boiling point or initial boiling point and boiling range	Not applicable for mixture.
Flammability	Not flammable
Lower and upper explosion limit/ flammability limit	Not applicable for solid
Flash point	Not applicable for solid
Auto-ignition temperature	Not applicable for solid
Decomposition temperature	Not determined for mixture



pH value
Kinematic viscosity
Solubility
Partition coefficient
n-octanol/water (log value)
Vapour pressure
Density and/or relative density
Relative vapor density
Particle characteristics

Metalaxyl: Stable to temperature up to 300°C
Propamocarb HCl: Stable to temperature up to 400°C
5.0-8.0
Not applicable for solid.
Form suspension in water.
Generally not applicable for mixture.

Not applicable for the solid mixture.
Not determined.
Not applicable for solid.
Min 98% through 44µm sieve.

9.2 Other information

Further safety related physical-chemical data are not known.

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

No reactivity under normal conditions.

10.2 Chemical stability

This product is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

10.3 Possibility of hazardous reactions

This product does not react or polymerize, releasing excess pressure or heat, or creating other hazardous conditions.

10.4 Conditions to avoid

Avoid fire, feed, food and beds of water.

10.5 Incompatible materials

Not compatible with strong oxidizing agents.

10.6 Hazardous decomposition products

When involves in a fire, maybe release oxides of carbon and nitrogen and chlorides and other toxic nitrogen compounds on combustion.

SECTION 11: TOXICOLOGICAL INFORMATION

Acute toxicity

Acute oral LD₅₀ for rats >3160 mg/kg (*Data From experimental report*)

Acute dermal LD₅₀ for rats >2150 mg/kg (*Data From experimental report*)

Metalaxyl: Acute inhalation LC₅₀ (4h) for rats >3.35mg/L (*Data From experimental report*)

Propamocarb HCl: Acute inhalation LC₅₀ (4 h) for rats >5.54



mg/l air. (BCPC, *e-The Pesticide Manual, Thirteenth Edition*)

Skin corrosion/irritation

Not irritant to skin (Rabbit) (*Data From experimental report*)

Serious eye damage/irritation

Slightly irritant to eyes. (Rabbit) (*Data From experimental report*)

Respiratory or skin sensitization

Not a skin sensitizer (Maximisation test) (*Data From experimental report*)

Germ cell mutagenicity

Metalaxyl: No genotoxic potential (*Data from EFSA Journal SANCO/10476/2010 – rev.1*)

Propamocarb HCl: Negative in Ames and micronucleus tests. (BCPC, *e-The Pesticide Manual, Thirteenth Edition*)

Carcinogenicity

Metalaxyl: Not carcinogenic (EFSA Journal SANCO/10476/2010 – rev.1)

Propamocarb HCl: No carcinogenic effects. (BCPC, *e-The Pesticide Manual, Thirteenth Edition*)

Reproductive toxicity

Metalaxyl: No reproductive effects at parentally toxic doses. (EFSA Journal SANCO/10476/2010 – rev.1)

Propamocarb HCl: No reproductive and developmental effects (BCPC, *e-The Pesticide Manual, Thirteenth Edition*)

STOT-single exposure

No available data.

STOT-repeated exposure

No available data.

Aspiration hazard

No available data.

Further information

No available data.

SECTION 12: ECOLOGICAL INFORMATION**12.1 Eco-toxicity****Birds**

Metalaxyl: Acute oral LD₅₀ for mallard ducks (8 d) 1466, Japanese quail (7 d) 923 mg/kg. LC₅₀ (8 d) for mallard ducks, Japanese quail, bobwhite quail >10000 mg/kg.

Propamocarb HCl: Acute oral LD₅₀ for bobwhite quail and mallard duck >1842 mg/kg.

Dietary LC₅₀ for bobwhite quail and mallard duck >4789 mg/kg.



- Fish** Metalaxyl: LC₅₀ (96 h) for rainbow trout, carp, and bluegill sunfish >100 mg/l.
Propamocarb HCl: LC₅₀ (96 h) for bluegill sunfish >92, for rainbow trout >99 mg/l.
- Daphnia** Metalaxyl: LC₅₀ (48 h) >28 mg/l.
Propamocarb HCl: LC₅₀ (48 h) 106 mg/l.
- Algae** Metalaxyl: IC₅₀ (5 d) for *Scenedesmus subspicatus* 33 mg/l.
Propamocarb HCl: ErC₅₀ (72 h) for *Selenastrum capricornutum* >85 mg/l; EbC₅₀ >120 mg/l.
- Bees** Metalaxyl: Acute toxicity LD₅₀ (48 h, contact) > 200µg /bee. Acute toxicity LC₅₀ (48h, oral) 269 mg/L (EFSA Journal SANCO/10476/2010 – rev.1)
Propamocarb HCl: LD₅₀ (oral) >84 µg/bee; (contact) >100 µg/bee.
- Worms** Metalaxyl: LC₅₀ (14 d) > 1000 mg a.s./kg substrate (EFSA Journal SANCO/10476/2010 – rev.1)
Propamocarb HCl: LC₅₀ (14 d) >660 mg/kg soil.

Above data is from BCPC, e-The Pesticide Manual, Thirteenth Edition

12.2 Persistence and degradability

In soil, DT₅₀ of Metalaxyl is 29 d (realistic range 10-40 d), Koc 70 ml/g (realistic range 30-300 ml/g). DT₅₀ in water 22-48 d. Photolytically stable in water and on soil surfaces.
Rapidly degraded in soil by microbial processes, following a brief lag phase, DT₅₀ <30 d, DT₉₀ <70 d. Propamocarb hydrochloride is retained in the upper soil layer (4-20 cm) and little is found in leachate. Stable in aqueous medium, but rapidly degraded by aquatic micro-organisms (up to 97% in 35 d). It is adsorbed onto sediment, but with limited desorption.

12.3 Bio-accumulative potential

Metalaxyl: An estimated BCF of 4 was calculated for metalaxyl(SRC), using log Kow of 1.65(1) and a regression-derived equation(2). According to a classification scheme(3), this BCF suggests the potential for bioconcentration in aquatic organisms is low(SRC). (Data from pubchem)
Propamocarb HCl: An estimated BCF of 54 suggests the potential for bioconcentration in aquatic organisms is low(Data from PPDB). K_{OW} logP = -1.21 (pH 7)

12.4 Mobility in soil

Metalaxyl: Based on a classification scheme(1), Koc values ranging from 30 to 284(2) indicate that metalaxyl is expected to have very high to moderate mobility in soil(SRC). (Data from pubchem)
Propamocarb hydrochloride is retained in the upper soil layer (4-20 cm) and little is found in leachate.

12.5 Other adverse effects

None



SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Product

The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.

13.2 Contaminated packaging

Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Do not burn, or use a cutting torch on, the empty drum.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO:

UN number: N/A

UN proper shipping name: GENERAL GOODS.

Transport hazard class: N/A

Packing group: N/A

Environmental hazards: Not a marine pollutant

Special precautions for user: None.

Transport in bulk according to IMO instruments: Not applicable

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the product in question

WHO-classification: III (Slightly hazardous)

This product is not subject to any prohibitions or restrictions in China.

SECTION 16: OTHER INFORMATION

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the PRODUCT AS SUCH. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produces formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.